R20

Code No.: 20ITC20

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.E. (IT, AI&DS) V Sem (Main & Backlog) Examination Dec 2023 – Jan 2024

Computer Networks

Time: 3 Hours **Max Marks: 60**

Note: Answer **ALL** questions from **Part-A & Part –B (Internal Choice)** at one place in the same order

Part - A
(5Q X 3M = 15 Marks)

			M	CO	BT
1		Outline the use of Cyclic Redundancy Check.	(3)	1	4
2		Explain briefly why both MAC and IP addressing schemes are required in the same network.	(3)	2	2
3		List the functions of Transport Layer.	(3)	3	2
4		How congestion occurs in a network.	(3)	4	3
5		List the Application Layer protocols.	(3)	5	2

Part – B
(5Q X 9M = 45 Marks)

			M	CO	BT
6	(a)	Discuss the TCP/IP reference model with neat sketch in detail.	(5)	1	2
	(b)	The message 11001001 is to be transmitted using the CRC polynomial $x^3 + 1$ to protect it from errors. What is the message that should be transmitted?	(4)	1	3
		(OR)			
7	(a)	Sixteen-bit messages are transmitted using a Hamming code. How many check bits are needed to ensure that the receiver can detect and correct single-bit errors? Show the bit pattern transmitted for the message 1101110000110101. Assume that odd parity is used in the Hamming code.	(5)	1	2
	(b)	Discuss about Carrier Sense Multiple access protocols in detail.	(4)	1	3
8	(a)	Explain Link state routing algorithm in detail.	(5)	2	3
	(b)	Why is IPV4 to IPV6 transition required?	(4)	2	2
		(OR)			
9	(a)	An IP datagram of size 1000 bytes arrives at a router. The router has to forward this packet on a link Whose MTU (Maximum Transmission Unit) is 100 bytes. Assume that the size of the IP header is 20 bytes. Compute the number of fragments that the IP datagram will be divided into for transmission.	(5)	2	3
	(b)	Change the following IPv4 addresses from binary notation to dotted-decimal notation. i) 10000001 00001011 00001011 11101111 ii) 11000001 10000011 00011011 11111111	(4)	2	2

10	(a)	Explain about Berkeley Sockets in detail	(5)	3	3
	(b)	Explain how TCP connection management is implemented.	(4)	3	4
		(OR)			
11	(a)	Explain how Multiplexing and Demultiplexing work in the transport layer.	(5)	3	3
	(b)	Discuss the purpose of transport layer in computer network.	(4)	3	2
12	(a)	Explain UDP header format in detail with neat sketch.	(5)	4	3
	(b)	Discuss UDP characteristics in detail.	(4)	4	2
		(OR)			
13	(a)	Discuss the role of Transmission Control Protocol (TCP) in providing a robust and stable communication framework in computer networks.	(5)	4	3
	(b)	Explain TCP Timer Management in detail.	(4)	4	2
14	(a)	Describe how SMTP protocol is used in E-mail applications.	(5)	5	2
	(b)	Write a short note on Domain Resource Records.	(4)	5	2
		(OR)			
15	(a)	Explain the process of web page loading the given page: https://www.cbit.ac.in/academic_post/department-of-computer-science-and-engineering/ Assume the IP address is unknown HTTP 1.1 is used, web page is available.	(5)	5	3
	(b)	Write a short note on WWW.	(4)	5	2

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